ZAGROSIA

ENGINEERING

Website Design & Development

Internship Presentation — BCIT NMDW

Morsal Mozneb

Operations & Web Dev

Farnaz Gholami

UX/UI Designer

May – July 2026

Meet the Team

M

Morsal Mozneb

Operations & Front-End Developer

Next.js · TypeScript · Tailwind · Framer Motion

F

Farnaz Gholami

UX/UI Designer

Figma · Wireframing · Prototyping · Design Systems

The Company

Zagrosia Engineering Inc. / i3 Building Science & Consulting

35+

Years of Expertise

3

Provincial Licences

P.Eng

BC · Alberta Saskatchewan

Founded by Seyed Hassan Mozneb, M.Sc., P.Eng., Zagrosia Engineering is a structural engineering firm serving developers, contractors, and building owners across Western Canada. The firm specializes in structural design, seismic assessment, building envelope, and racking systems — while pushing the frontier of AI automation in engineering practice.

INTRO

The Project

DURATION

9 weeks (May 10 – July 5, 2026)

HOURS

340+ combined hours

PAGES BUILT

8 full pages

DELIVERABLE

Live production website

8 Pages

01 Home
02 About
03 Services
04 Projects
05 AI Lab
06 Academy
07 Blog
08 Contact

Internship Experience

9 weeks · May – July 2026

WEEK 1

Week 1 — Onboarding

Email Infrastructure

Set up Microsoft 365 for the firm — info@zagrosia.ca, professional email accounts. Configured DNS records via GoDaddy.

LinkedIn Business Page

Created and launched the Zagrosia Engineering LinkedIn company page with branded content and company overview.

Competitor Research

Analyzed 5+ structural engineering firm websites across BC. Identified gaps in digital presence and UX patterns to improve on.

Target Client Definition

Mapped ideal clients: project owners, developers, contractors in Western Canada. Informed navigation and content strategy.

WEEK 1

UX Research & Planning

Farnaz — Week 1

Sitemap Planning

Defined site architecture: 8 primary pages with clear hierarchy and user journeys mapped for each audience type.

User Flow Mapping

Created detailed user flows for key conversion paths — from landing to contact form, from Academy to exam prep.

Low-Fi Wireframes

Completed lo-fi wireframes for Home, About, and Contact pages. Fast iteration in Figma to test layout logic before committing to design.

Business Card Design

Designed professional business cards for the firm's P.Eng — aligned with brand identity and print specifications.

WEEKS 2–3

Weeks 2–3: Digital Operations

Morsal — establishing the company's digital presence beyond the website

01

Content Strategy

Developed a content framework for the website and LinkedIn — defining tone, messaging hierarchy, and key differentiators for a structural engineering audience.

02

Brand Documentation

Compiled brand guidelines alongside Farnaz's design system — colors, typography, logo usage, and image standards for consistent application across all channels.

03

SEO Foundation

Researched keyword opportunities for structural engineering in BC and Alberta. Set metadata strategy for each of the 8 website pages.

WEEKS 2–4

From Lo-Fi to Hi-Fi

Farnaz — Figma design process, weeks 2 through 4

STAGE 1

Low-Fidelity

Weeks 1–2

- Grayscale wireframes for all 8 pages
- Layout and information hierarchy validated
- Fast iteration — 3–4 versions per key page

STAGE 2

Mid-Fidelity

Week 3

- Added real content and spacing
- Typography hierarchy locked
- Component patterns established

STAGE 3

High-Fidelity

Weeks 4–5

- Full color, imagery, and micro-copy
- Interactive Figma prototype
- Client review-ready deliverable

Client Review Meeting

Week 5 — Presenting the Hi-Fi Figma prototype to Homayoun Mozneb, P.Eng.

"The first time the client saw all 8 pages together — live, interactive, and branded. A 2-hour session that shaped the final product."

Key Feedback Incorporated

- Refine hero section typography
- Add P.Eng credential badges
- Strengthen AI Lab section presence
- Expand services detail copy

Takeaway: Real client feedback revealed priorities that specs alone never could. The meeting taught us to design for communication, not just aesthetics.

Development Begins

Morsal — Week 4: All 8 pages built in a single intense development sprint

AI-Assisted Development Stack

Codex

AI code generation for boilerplate and component scaffolding

Claude Code

Complex logic, refactoring, and multi-file coordination

V0 by Vercel

UI component generation from design descriptions

GitHub Copilot

Inline completions while writing TypeScript

Pages Completed in Week 4

- ✓ Home
- ✓ AI Lab
- ✓ About
- ✓ Academy
- ✓ Services
- ✓ Blog
- ✓ Projects
- ✓ Contact

Refinement & Polish

Weeks 6–9 — Desktop layouts, animations, QA, and launch

Wk 6	Wk 7	Wk 8	Wk 9
Desktop Layouts	Card Design & Shadows	Scroll Animations	QA & Launch

Desktop-First Overhaul

Rebuilt layouts for 1280px+ screens: wider containers, multi-column grids, larger typography scales, and refined spacing ratios.

Framer Motion Animations

Added scroll-triggered entrance animations to every section. Custom spring physics for stagger effects, blur fades, and slide-ins using Framer Motion.

Magnetic CTA Buttons

Built a custom CtaButton component with three layered effects: magnetic cursor pull (spring physics), shimmer light sweep on hover, and spring press on click.

QA & Accessibility

Cross-browser and cross-device testing. Implemented prefers-reduced-motion support — all animations disabled for users who opt out.

Showcase Your Work

zagrosia.ca — 8 pages, live in production

OVERVIEW

zagrosia.ca — At a Glance

8	**35+**	**100%**	**Live**
Pages	**Components**	**Responsive**	**Production**
Built from scratch	React / Next.js	Mobile-first	Deployed on Netlify

Page Structure

- **Home** — Hero, Stats, Services teaser, AI Lab, EGBC Prep, Blog
- **About** — Company story, values, team, milestones
- **Services** — 6 structural engineering services
- **Projects** — Portfolio with filter by project type
- **AI Lab** — AI & automation tools and demos
- **Academy** — EGBC Exam Prep with course cards
- **Blog** — Engineering insights and updates
- **Contact** — Contact form + company info

HOME

Home Page

The first impression — animated hero, social proof, and all major platform sections

Hero Section	Full-viewport hero with animated headline, "Engineered to Solve" tagline, and two magnetic CTA buttons. Background rotates between structural engineering imagery.
Engineering Stats	35+ years, P.Eng BC/AB/SK credentials, project count — displayed as animated count-up numbers triggered on scroll.
AI Lab Preview	Dark-themed section highlighting AI & automation capabilities. 500+ videos, 150+ subjects in the AI education library.
EGBC Exam Prep	Teaser for the Academy section — exam preparation platform for engineering licensure candidates in BC.

Services Page

Six structural engineering services — each with detailed capability descriptions

01

Structural Design

Residential & commercial building systems, foundations, lateral loads

02

Seismic Assessment

Seismic vulnerability studies and retrofit design per BC Building Code

03

Rehabilitation

Structural assessment, damage investigation, repair specifications

04

Racking Systems

Industrial racking and mezzanine structural review and stamp

05

Building Envelope

Enclosure assessment, moisture investigations, cladding design

06

Specialty Reviews

Peer review, letter of assurance, as-built review services

Projects Portfolio

Showcasing real completed engineering projects with filterable card layout

Card Layout

Each project displayed as a rich card with project type, location, scope, and featured image.

Filter System

Users filter by project type — Residential, Commercial, Industrial, Seismic — for targeted browsing.

Animations

Cards animate in with stagger effect on scroll. Hover reveals overlay with project details.

Project Hero

Dedicated hero section with animated stat callouts and project category breakdown.

AI & Automation Lab

Where structural engineering meets artificial intelligence

500+
Educational Videos

150+
Engineering Subjects

200+
Total Views

Platform Capabilities

- Custom Python scripts for structural calculation automation
- VBA macros for Excel-based engineering workflows
- AI-powered educational platform for engineering topics
- Automation demos for common engineering firm tasks

Academy — EGBC Exam Prep

A structured learning platform for engineering candidates pursuing BC licensure

What It Offers

- Structured study roadmap for EGBC PPE exam
- Video lessons covering exam-critical concepts
- Practice questions and self-assessment tools
- Downloadable resources and formula sheets
- Progress tracking and study milestones

Who It's For

Engineering Graduates

Preparing for the BC PPE exam after completing their engineering degree.

International Engineers

Internationally trained engineers seeking BC licensure through EGBC.

Exam Retakers

Engineers who need to strengthen specific topic areas before retaking.

MORE PAGES

Blog, About & Contact

Blog

- Engineering insights and firm updates
- Card-based layout with category tags
- Hover animations with read-more CTA
- SEO-optimized metadata per post

About

- Company story and 35+ year history
- Desktop: split two-column layout
- Mobile: stacked with scroll reveals
- Team section and values statement

Contact

- Animated contact form (validated)
- Company phone, email, and address
- Magnetic CTA submit button
- Success/error state feedback

Technology Stack

Every tool chosen for a reason

FRAMEWORK

Next.js 15
App Router, SSR, file-based routing

TypeScript
Type safety across all components

STYLING

Tailwind CSS
Utility-first, mobile-first design

Framer Motion
Physics-based scroll animations

DESIGN

Figma
Full design system + prototypes

React Icons
Consistent iconography across pages

DEVOPS

GitHub
Version control, CI/CD trigger

Netlify
Auto-deploy on push to main

AI TOOLS

Codex / Claude Code
AI-assisted development

V0 by Vercel
UI component generation

Signature Animations

The interactions that make zagrosia.ca feel alive

Scroll-Triggered Reveals

Framer Motion · whileInView

Every section animates in as it enters the viewport. Four effects: fade-up, fade-left, blur-in, and scale-up. Staggered delays create a cascade that guides the eye down the page.

Magnetic CTA Buttons

useMotionValue · useSpring

Buttons drift toward the cursor using spring physics. On hover, a light shimmer sweeps diagonally across. On click, the button springs down to 93% scale and bounces back — like pressing a physical button.

Count-Up Numbers

useInView · requestAnimationFrame

Statistics (35+ years, 500+ videos) count up from zero when scrolled into view. Each number has its own duration and triggers only once — reinforcing the credibility of each claim.

Accessibility First

useReducedMotion()

All animations check prefers-reduced-motion. If the OS accessibility setting is on, every animation silently disables — same components, zero motion. Inclusivity built in at the component level.

DESIGN

Design System

Farnaz — the visual foundation that makes every page consistent

COLOR PALETTE

#0052A5
Navy — Primary

#FCFCFC
White — Background

#0D0D0D
Black — Dark sections

#94B8DC
Blue — Accent

TYPOGRAPHY

Role	Size	Weight	Used For
Display / H1	40–52px	Medium 500	Hero headlines
Section / H2	24–32px	Normal 400	Section titles
Body	12–16px	Regular	Content paragraphs
Caption / Label	8–11px	Light · tracking	Tags, metadata

What We Learned

Technical skills, collaboration, and real-world growth

MORSAL

Technical Growth — Morsal

Next.js 15 App Router

Built a production app using the full App Router paradigm — server vs client components, route groups, metadata API, and image optimization. Not just following tutorials: architecting real pages.

TypeScript in Practice

Went from avoiding TypeScript to relying on it. Interfaces, union types, and type guards caught real bugs before they hit production. Type safety became a workflow, not a chore.

AI as a Development Tool

Learned to use Codex, Claude Code, and V0 as precision instruments — not magic. The skill is knowing what to prompt, how to evaluate the output, and when to throw it away.

Git & CI/CD Pipeline

Managed the full GitHub → Netlify deployment pipeline. Diagnosed and fixed real deployment failures (pnpm lockfile conflicts, plugin config). Production-level DevOps, not sandbox practice.

Animation with Framer Motion

Implemented spring physics, scroll triggers, and gesture-based animations from scratch. Built the custom CtaButton component with three layered motion effects and full accessibility support.

FARNAZ

Design Growth — Farnaz

Research-Driven Design

Learned to validate every design decision with research — competitor analysis, user flows, and sitemap logic before a single frame was opened in Figma. Design without research is guesswork.

Figma Prototyping

Advanced prototype techniques: interactive components, smart animate transitions, and reusable design tokens. The client could click through the prototype as if it were a real website.

Client Communication

Presenting design work to a real client is different from a school critique. Learned to frame decisions in business terms — and to listen for the real need behind feedback.

Responsive Design Principles

Designed for mobile first, then adapted to desktop. Maintaining visual hierarchy across breakpoints requires a systematic approach to spacing and type scale.

Design-to-Dev Handoff

Worked directly with Morsal to ensure designs were implementable. Tighter collaboration produces better outcomes than a clean handoff.

Designer + Developer Collaboration

What we learned from working together on a real product

Research & Wireframes	Hi-Fi Prototype & Client Review	Development & Animation	QA & Launch
Farnaz	Both	Morsal	Both

What surprised us:

Design decisions that seem obvious in Figma are often the hardest to implement — and implementing them reveals better alternatives.

Biggest challenge:

Keeping design and code in sync across 9 weeks without a formal handoff tool. We solved it with frequent syncs and shared Figma access.

What we'd do differently:

Establish a component library in Figma that maps 1:1 to the React components from the start. Naming mismatches caused friction mid-project.

INSIGHTS

Key Insights

01

AI tools don't replace judgment — they amplify it

Every AI-generated component still required architectural thinking, design evaluation, and code review. The bottleneck shifted from 'can I write this?' to 'is this the right thing to write?'

02

Real clients operate differently from academic briefs

The brief changed. Priorities shifted. A 2-hour review session reshaped the product more than weeks of planning. Adaptability and clear communication became technical skills.

03

Polish is where the professional gap lives

The gap between a working website and a professional product is filled with animation timing, spacing consistency, type scale, and accessibility. These aren't nice-to-haves. They're the product.

04

Design and code are two halves of one discipline

The best decisions happened when designer and developer were in the same conversation. Silos produce handoff friction. Collaboration produces better products.

Looking Ahead

From internship to career

LIVE

It's Live.

zagrosia.ca

Real Client

A practicing P.Eng with 35+ years of experience now has a digital presence that reflects the scale and quality of their firm.

Real Stack

Next.js 15 · TypeScript · Tailwind CSS · Framer Motion · GitHub → Netlify CI/CD. Production infrastructure, not a classroom project.

Real Handoff

The site was delivered to the client with documentation, admin access, and a deployment pipeline that auto-updates on every git push.

What's Next

Morsal

- Continue contributing to zagrosia.ca — CMS integration, blog automation, and advanced AI features
- Pursue full-stack development roles in BC tech — Next.js, TypeScript, and animation-forward products
- Apply AI-assisted development workflow to future freelance and agency projects

Farnaz

- Build out the Zagrosia design system as a reusable Figma library for future firm materials
- Pursue UX/UI design roles at product companies and agencies in Metro Vancouver
- Continue developing expertise in interaction design and design-to-dev handoff workflows

Thank You

Questions & Discussion

Morsal Mozneb · Farnaz Gholami

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